

VET436 – Cardiac Diseases

Much of the information relating to cardiac disease in small mammals results from collation of sporadic case reports and case series. Pharmacologic, echocardiographic, electrocardiographic, and radiographic studies continue to be published and are helping to establish drug dosages and reference intervals for diagnostic imaging. The pharmacotherapy of heart disease in small mammals has been extrapolated from that of the dog and cat, as much of the pathophysiology is the same. These notes briefly summarize the diagnosis and treatment of cardiovascular diseases in small mammals with a focus on rabbits, guinea pigs, and ferrets.

Cardiac Anatomy

The small mammal heart is nearly identical in structure to that of dogs and cats. The mammalian heart has 4 chambers. The atria and ventricles are separated by atrioventricular valves, the tricuspid valve on the right and the mitral valve on the left. The pulmonary valve (3 cusps) sits between the right ventricle and pulmonary artery. The aortic valve (3 cusps) sits between the left ventricle and aorta. A cranial and caudal vena cava return venous blood to the right atrium. In most small mammal species, there are two branches at the aortic arch (left subclavian and a brachiocephalic trunk in rabbits and guinea pigs). The pericardial sac surrounds the heart and proximal great vessels and is made up of two layers, visceral (serous) and parietal (fibrous), that create a potential space (pericardial space) that contains a very small amount of lubricating fluid. Some interesting (but clinically insignificant) differences include the following: the tricuspid valve of the rabbit is composed of two cusps; rabbits (and rats) often have persistence of the left cranial vena cava, which drains into the right atrium via the coronary sinus.

History

Obtain a complete history, including husbandry practices, diet, medications and supplements, and past and present illnesses. Historical findings in animals with heart disease include reduced activity level, respiratory abnormalities, lethargy, weakness, syncope, anorexia, or weight loss.

Physical Examination

It is important to observe small mammals in their transport carrier prior to any handling. Quantify the respiratory rate, observe the respiratory character and the animal's behavior. Dyspnea is an emergency and is associated with increased work of breathing, nasal flaring, behavioral signs of distress, and/or orthopnea. If the animal is working hard to breathe, yet not showing signs of distress (e.g., is interested in their environment), this usually less emergent and represents successful adaptation to respiratory disease. Cyanosis in the context of abnormal respiratory signs usually indicates hypoxemia from airway obstruction or pleural space disease. Hypoxemia due to heart failure usually leads to a greying of the mucous membranes especially with stress or exertion. Whereas fast, shallow breathing may be associated with 'stiff lungs' (edema, fibrosis, disseminated neoplasia), vigorous large excursion of the chest and abdomen are more suggestive of pleural space disease. Recruitment of the abdominal muscles for breathing leads to contraction of the abdomen in exhalation and is usually present when there is resistance to exhalation (especially small airway collapse).

When you are ready to handle the animal, do so in a quiet room using a table with a non-slip surface. The cardiovascular exam includes assessment of the mucous membranes (usually light pink), capillary refill time (normal is less than 2 seconds), peripheral arterial pulse quality, and thoracic and cardiac auscultation. Lung sounds are usually characterized as increased, decreased, absent, or normal. Absent lung sounds in animal with tachypnea or dyspnea should raise concern for pleural effusion. Conversely, absent lung sounds in a eupneic animal with normal effort and cardiovascular parameters may be normal. Muffled heart sounds are supportive of pericardial effusion. Increased lung sounds are not specific and may be caused by many kinds of respiratory diseases. Adventitial sounds such as crackles are most supportive of either edema or fibrosis.

Like dogs and cats, only two of the four heart sounds (S1 and S2) are heard in normal animals. The S1 and S2 sounds frame systole and represent closure of the mitral/tricuspid and aortic/pulmonary valves respectively and are generated by the oscillation of the blood column. The presence of a third (gallop) sound (S3 or S4, which are indistinguishable at the high rates of small mammal hearts) indicates diastolic dysfunction and is a specific finding for heart disease in the rabbit and ferret.

Physical exam findings that support specific abnormalities

1. Left sided systolic heart murmur
 - a. Mitral valve insufficiency (degeneration, congenital dysplasia, functional due to left ventricular and annular dilation [DCM], or infectious endocarditis)
 - b. Aortic valve stenosis (degeneration, congenital dysplasia, infectious endocarditis)
 - c. Left ventricular outflow tract obstruction (HCM)
2. Right sided systolic heart murmur
 - a. Tricuspid valve insufficiency (degeneration, congenital dysplasia, functional due to pulmonary hypertension or dilation and resultant remodeling, or infectious endocarditis)
 - b. Left to right shunting ventricular septal defect
3. Left sided continuous heart murmur
 - a. Patent ductus arteriosus

Calculate the heart rate, determine if it is normal for the species (reference intervals present in Power Point notes), and characterize the rhythm as regular or irregular. Any abnormalities in heart rate or rhythm are indication for an electrocardiogram (see below).

Heart diseases that lead to a reduction in cardiac output will lead to prolonged capillary refill time, a reduction in peripheral arterial pulse quality, and/or a cool periphery. In addition to tachypnea or dyspnea, signs of left-sided heart failure also include inspiratory crackles and, in some species, cough. Signs of right-sided heart failure include jugular venous distention and/or pulsation, signs of pleural effusion (see above), hepatomegaly, and signs of abdominal effusion (abdominal distention, ballotable fluid wave).

For a list of the most common cardiac diseases in rabbits, rodents, and ferrets, please see the Power Point slide notes.

Cardiac Diagnostics

The same diagnostic tools used in larger mammals are used in small mammals.

Point of Care Ultrasound (POCUS)

Point of care ultrasound is a useful cage side tool to diagnose cavitory effusions. Lung ultrasound can be used screen for B-lines, which are nonspecific and can be caused by edema (cardiogenic or non-cardiogenic), fibrosis, or other interstitial infiltrates. The focused cardiac exam can be used to roughly evaluate left ventricular size and systolic function and left atrial size (using the left atrium to aortic ratio; a ratio > 2 in the face of respiratory distress is highly supportive of congestive heart failure).

Survey Radiographs

Along with POCUS, survey radiographs are accessible to the general practitioner and provide useful information about the cardiac shape and size, pulmonary vasculature, and pulmonary parenchyma. Normal vertebral heart scale/score (VHS) reference intervals exist for some species of small mammals. Using the right lateral projection, the long axis of the heart is measured from the ventral aspect of the carina to the distal left ventricular apex (excluding any fat opacity) and the short axis is measured perpendicular to the long axis line at the level of the caudal vena cava. These two measurements are positioned over the thoracic vertebrae starting at the beginning of the body of T4 (or T5 for ferrets). The number of vertebrae spanned by these two lines are then added.

In the ferret, the heart may have the appearance that it is floating, this is a normal variation and in the eupneic animal should not be confused with a pneumothorax. In rabbits, the cranial margin of the heart may be obscured by fat, and it is ideal to pull the thoracic limbs forward to improve visualization of structures in the cranial thorax.

Radiographic Signs of Cardiac Chamber Enlargement

Left atrial enlargement leads to straightening to convexity of the caudal-dorsal silhouette and ultimately to elevation of the carina and left main-stem bronchus on the lateral view. Left atrial enlargement may lead to a widening/divergence of the main stem bronchi on the DV or VD view. Left atrial appendage (auricle) dilation is best seen on the DV or VD view as a bulge at the 2 to 3 o'clock position. Left ventricular dilation leads to an increase in the height of the heart and a more acute angle between the trachea and the spine on the lateral view. The left ventricular apex may also appear rounder on the DV or VD view. Concentric hypertrophy of the left ventricular may not be radiographically apparent or may cause a slight convexity of the caudal border of the silhouette on the lateral view. This change cannot be easily differentiated from dilation, however, and echocardiography is best to elucidate the type of left ventricular remodeling.

Enlargement of the cranial-dorsal silhouette on the lateral view may be due to right atrial enlargement or pulmonary artery or aortic dilation. The DV or VD view may help determine which of these structures is enlarged (aortic enlargement appears at approximately 12 o'clock and pulmonary artery enlargement at approximately 1 o'clock). Right atrial enlargement on this view may appear as rounding of the right heart border from the 9 o'clock to 11 o'clock position. Right ventricular dilation leads to increased sternal contact on the lateral view, though the amount of normal contact varies amongst species. On the DV or VD view, right ventricular

enlargement may lead to the reversed D appearance of the silhouette. Right ventricular hypertrophy may cause lifting of the left ventricular apex dorsally on the lateral view. Generalized cardiomegaly may arise from enlargement of all four chambers. Pericardial effusion can also lead to an appearance of generalized enlargement.

Pulmonary Vasculature

The pulmonary artery and veins are separated by a bronchus with veins being ventral on the lateral view and central on the DV or VD view in relation to the bronchus. The two pairs of cranial lobar vessels are best seen on the left lateral view whereas the DV view is superior to the VD view for visualization of the caudal lobar vasculature. The artery and vein should be the same size and are often compared to one another. The width of the vessels can also be compared to the width of the ribs (the fourth rib as it crosses the T4 vertebra on the lateral view and the 9th rib on the DV or VD view). The widths should be the same and a wider vessel supports dilation and a narrower vessel a reduced size.

Pulmonary Parenchyma and Pleural Space Abnormalities in Heart Disease

Cardiogenic pulmonary edema usually manifests as an unstructured interstitial infiltrate that tends to be most dense in the perihilar and caudal lung fields. With increased severity, the edema becomes alveolar, leading to an alveolar pattern in the most severely affected lung fields. The concurrent finding of left-atrial enlargement and pulmonary venous dilation support a diagnosis of cardiogenic pulmonary edema. Pulmonary venous enlargement may be absent in patients receiving furosemide therapy. Pleural effusion is usually present with right-sided congestive heart failure yet also occurs along with pulmonary edema in bi-ventricular congestive heart failure. Pleural effusion leads to a soft tissue opacity that causes widening of the interlobar fissures, retraction of the lung from the thoracic wall, and when more severe, obscures the cardiac silhouette and diaphragm. Left-sided heart failure can lead to mild to moderate post-capillary pulmonary hypertension, though pulmonary artery size often remains normal despite this. More severe pulmonary arterial dilation and tortuosity should increase the suspicion of a primary cause of pulmonary hypertension. The pulmonary over circulation resulting from congenital shunts such as a left-to-right shunting patent ductus arteriosus or ventricular septal defect can lead to dilation of both pulmonary arteries and veins, leading to a vascular pattern.

Electrocardiography

The indications for an electrocardiogram (EKG) are an abnormal heart rate and/or rhythm. In small patients with high heart rates, ideal paper speeds are 50 to 100 mm/second and ideal amplitude calibration is 20 mm/mV. Whereas physiologic arrhythmias (respiratory sinus arrhythmia, Mobitz Type I AV block) may occur in the calm, healthy ferret, they are not expected in the healthy rabbit or guinea pig. Clip electrodes are usually able to be used in small mammals. Gel soaked, thin gauze can be applied between the skin and the clip to protect the skin. Alcohol is irritating to ferrets and only electrode gel is used. Sternal recumbency is usually adequate to get a rhythm strip and evaluate intervals (PR/PQ, QT) and durations (P, QRS). A hanging technique is often used in ferrets to keep them still while recording the EKG. Care should be taken to ensure there is good lead contact and minimal motion to allow for a smooth, artifact free baseline. The lead configuration is the same as in the dog and cat.

Each P-QRS-T represents one cardiac cycle. Depolarization of the sinoatrial node (SAN) is not seen on the surface EKG, yet triggers depolarization (and contraction) of the atrial myocardium which is evidenced by a P wave. Normally the P wave is positive in leads II, III, and aVF. This is followed by the PQ interval, as the impulse travels through the atrioventricular node (AVN). The impulse then arrives at the ventricles and causes depolarization and contraction – leading to generation of the QRS complex. The R wave is usually positive in leads II, III, and aVF. Repolarization of the ventricles is represented by the T wave. The T wave may be positive, negative, or biphasic.

Average HR determination

- 25 mm/s: Count number of QRS complexes in 6 seconds (30 big boxes) & multiply by 10
- 50 mm/s: Count number of QRS complexes in 3 seconds (30 big boxes) & multiply by 20
- A Bic® pen with the cap on is ~15cm (30 big boxes) long
- At 25mm/s – count # of complexes in pen, then x 10 (pen times 10)
- At 50mm/s – count # of complexes in pen, then x 20 (pen times 20)

Instantaneous HR determination

- Method 1:
 - At 25mm/s – 1500/#small boxes R-R
 - At 25mm/s – 300/#big boxes R-R
 - At 50mm/s – 3000/#small boxes R-R
 - At 50mm/s – 600/#big boxes R-R
- Method 2: 60/RR interval (in seconds)

Echocardiography

Echocardiography characterizes the structure and function of the heart and usually allows for the definitive diagnosis of heart disease. Multiple examples will be shown in class.

Heart Failure

Definition of Heart Failure

The pathologic state of heart failure occurs when the heart's ability to eject and/or receive blood is impaired. The heart must pump blood with enough force to effectively perfuse metabolizing tissues. Similarly, the heart needs to receive blood (from systemic veins on the right side and pulmonary veins on the left side) to effectively drain these capillary beds. When the heart is unable to efficiently pump blood to metabolizing tissues – the animal will develop signs of poor cardiac output (exercise intolerance, metabolic disturbances) and this is called low output failure. When the heart is unable to efficiently receive blood – venous and capillary pressures will rise, and the animal will develop signs of congestion (pulmonary edema, ascites, or pleural effusion) and this is called congestive heart failure (CHF). In many cases – both congestive and low-output heart failure are present. Right-sided CHF is characterized by increased systemic venous pressures and cavitory effusions (ascites, pleural and pericardial effusion). Left-sided CHF is characterized by increased pulmonary venous pressures and pulmonary edema (and sometimes pleural effusion in the cat). Bi-ventricular heart failure occurs in animals with both left and right-sided heart disease and is characterized by the presence of pulmonary edema and a cavitory effusion (usually ascites and/or pleural effusion).

Decreased cardiac output and inadequate perfusion resulting from heart disease ‘triggers’ compensatory mechanisms that include neurohormonal activation (sympathetic nervous system and the renin-angiotensin-aldosterone system) and cardiac remodeling (hypertrophy). Activation of these systems is another hallmark of heart failure. Neurohormonal activation favors a state of water retention and vasoconstriction. Myocardial remodeling (primarily hypertrophy) and the effects of neurohormonal activation allows for compensation (the maintenance of cardiac output) in the short to mid-term. Chronic neurohormonal activation, however, contributes to pathologic remodeling (i.e. myocardial fibrosis) and perpetuation of the syndrome of heart failure.

The inciting cause(s) of heart failure can be categorized anatomically (myocardium, endocardium/valves, conduction system, pericardium, and vascular system) and physiologically (systolic failure, diastolic failure, arrhythmia, hemodynamic [volume or pressure] overload). In many cases, more than one category is involved and whereas there may have been one inciting cause, additional factors may contribute to progression. The development arrhythmia may complicate any heart disease and lead to further reduction in cardiac output due to loss of atrioventricular synchrony and either too rapid or too slow ventricular response rates. Although the pharmacological toolkit for treating heart failure is relatively small, the art and challenge lies in designing a regimen that fits each animal’s unique clinical picture and changes as their disease evolves.

Therapy of Heart Failure

The approach to the management of heart failure depends on the underlying disease, its severity, and the type of signs present. The sections below outline typical drugs used in the management of heart failure. In addition to these drugs, oxygen therapy and mechanical removal of any cavitory effusions (thoracocentesis, abdominocentesis) are also important in the stabilization of patients with heart failure. Additionally, sedation may be used to reduce stress in anxious patients. Dosages are not emphasized here. Please see the suggested references in the Power Point slide notes.

Loop Diuretics (Furosemide)

Loop diuretics have long been the cornerstone in the management of congestive signs. The loop diuretic furosemide (Lasix) is by far the most used diuretic. It’s rapid onset of action makes it indispensable in the emergency treatment of cardiogenic pulmonary edema. It does, however, have a relatively short duration of action, which often necessitates repeated bolus dosing in the emergency setting. Usually, a bolus of furosemide at 1 to 2 mg/kg is given intravenously with diuresis expected within 30 to 45 minutes. Boluses are repeated as needed based on respiratory rate/effort and patient demeanor. The total dosage of furosemide and duration of hospitalization will of course depend on the severity of CHF. Chronic therapy with furosemide is ideally titrated to the lowest dosage that controls the patient’s congestion. Increasing furosemide dosages are required over time. When using any diuretic, it is important to evaluate renal function and electrolytes prior to and 7 to 10 days after initiation or dose escalation. When titrating a patient’s torsemide dosage or in a patient with kidney disease this monitoring may need to be more frequent.

Vasodilators (Nitroglycerin and ACE-Inhibitors)

Nitroglycerin (a venodilator) is used in emergency situations to reduce preload and pulmonary edema (2% ointment applied topically to inner pinna). Angiotensin converting enzyme inhibitors (enalapril, benazepril, ramipril) counteract the maladaptive and chronic activation of the renin angiotensin aldosterone system and the attendant sodium and water retention, vasoconstriction, and pathologic remodeling. Whether these drugs impact heart failure outcomes in small mammals remains to be seen. Dosages for several small mammal species have been reported.

Inotropic Therapy (Pimobendan)

Pimobendan is a calcium sensitizing/phosphodiesterase III inhibiting inotrope, better termed an inodilator (inotrope and mixed vasodilator). Pimobendan sensitizes the troponin C complex to calcium and enhances actin and myosin interaction. It is also a phosphodiesterase III inhibitor, which leads to mild peripheral arterial vasodilation (afterload reduction) and a slight increase in myocardial cell calcium (improved inotropy). Whether pimobendan impacts heart failure outcomes in small mammals remains to be seen. Dosages for several small mammal species have been reported.

Antiarrhythmic Therapy

As with antiarrhythmic therapy in the dog and cat, much of our drug choices are anecdotal and based on experience, and best guesses as to the electrophysiologic basis of the arrhythmia(s). Except for atrioventricular (AV) block in ferrets most arrhythmias in small mammals are tachyarrhythmias. The decision to treat usually depends on the frequency and rate (sustained, high-rate arrhythmias are more likely to cause clinical signs and hurt cardiac function and output) and malignancy (rapid and/or polymorphic ventricular tachycardia may lead to torsades de pointe and ventricular fibrillation) of the arrhythmias.

Atrial fibrillation in small mammals occurs secondary to underlying heart disease and resultant atrial remodeling. The treatment of atrial fibrillation is aimed at slowing the ventricular response rate rather than converting the patient back to sinus rhythm. Diltiazem is usually initiated first, and digoxin is added if rate control is inadequate with diltiazem alone. Ventricular tachycardia is usually treated with lidocaine in the emergency setting. When administering lidocaine, monitor for gastrointestinal (nausea, vomiting) and CNS (muscle twitching, convulsions) side effects. Sotalol is used as chronic oral therapy for malignant ventricular arrhythmias. Supraventricular tachycardias are often treated with either sotalol or diltiazem (e.g., try one first and switch to the other if it doesn't work, or use them in combination!).

In ferrets, low grade AV block (occasional 2:1 AV block, especially Mobitz Type I) does not cause clinical signs, is often found incidentally, and usually does not warrant therapy. These low-grade AV blocks, however, should be monitored. High grade second degree or complete (third degree) AV block usually warrants further investigation and therapy. An atropine response test can be used to determine if vagal influence is playing a role in high grade or complete AV block. A positive response to atropine suggests that sympathomimetic therapy may help reduce the severity of the AV block. AV block that is not responsive to atropine and has a ventricular rate less than 80bpm is likely to cause clinical signs (syncope) and heart failure (usually right-sided heart failure). Epicardial pacemaker implantation is usually required to palliate this rhythm and prevent heart failure and sudden death.